

3(a)(i)	gas for which $pV \propto T$	M1
	where T is thermodynamic temperature	A1
3(a)(ii)	no intermolecular forces	B1
	(so) potential energy is zero	B1
3(b)(i)	$pV = NkT$	C1
	$N = (2.0 \times 10^5 \times 0.26) / (1.38 \times 10^{-23} \times 290)$	A1
	$= 1.3 \times 10^{25}$	
3(b)(ii)	$E_K = (3/2) kT$	C1
	$E_K = (3/2) \times 1.38 \times 10^{-23} \times 290$	A1
	$= 6.0 \times 10^{-21} \text{ J}$	
3(b)(iii)	internal energy = total KE + PE of molecules or PE = 0 so internal energy = total KE of molecules	B1
	internal energy = $1.3 \times 10^{25} \times 6.0 \times 10^{-21}$	A1
	$= 7.8 \times 10^4 \text{ J}$	
3(c)	straight line with positive gradient	B1
	line passing through the origin	B1